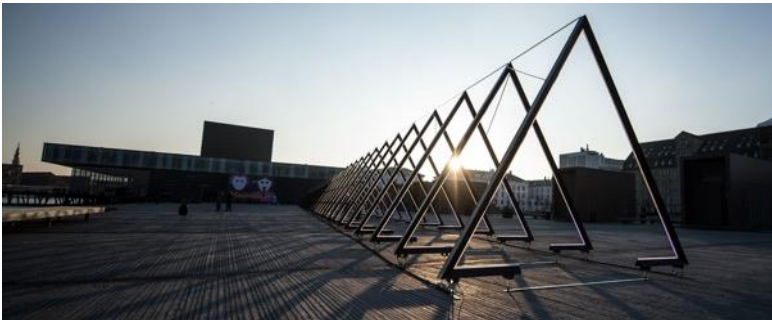
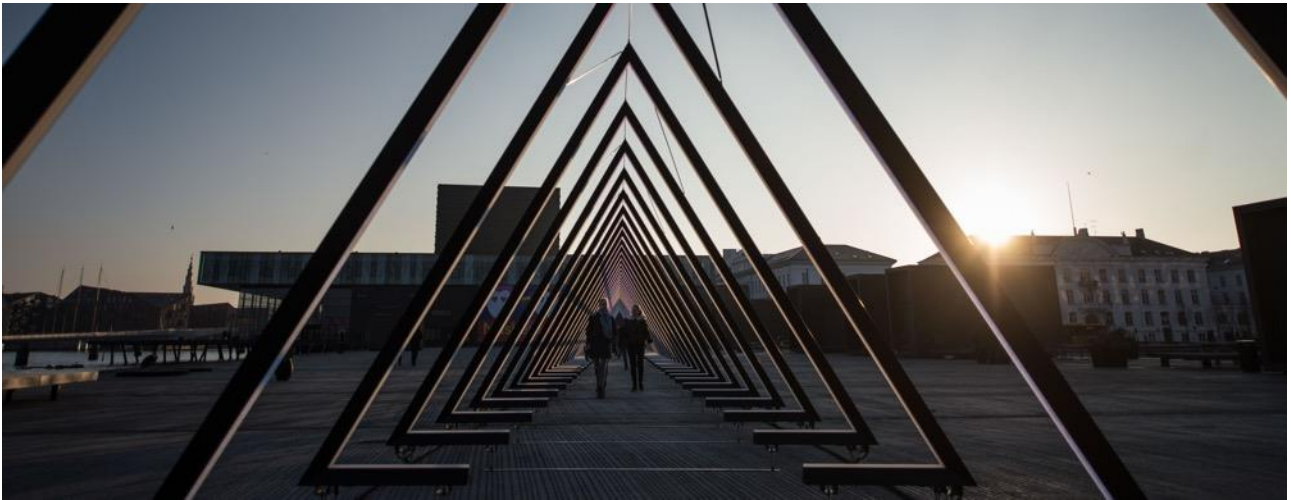




THE WAVE by VERTIGO

Technical rider





Description

"The Wave" consists of 40 four-meter-high interactive triangular light and sound gates.

Each gate reacts with light and sound to the visitors' movements through the work.

The public will together co-create a series of unique lighting and audio events, as their interaction with the work forms a constant change of the installation during the whole exhibition period.

The composition of light and sound consist of 3 moods, which are lasting for 3 minutes each running in a 9-minute loop. Usually people walk back and forth thru the gates a couple of times and gather in small groups to observe the piece.

"The Wave" is run by a mac computer driving 2 separate networks. One for control of the 350 meters of LED based on the ArtNet protocol, and another to distribute sound to the 80 speakers via the Dante protocol.

In case of longer exhibition where an assistant from Vertigo cannot be present during the whole period. A workshop will be held where we will train local staff to fail seek and maintain the piece, see description below.

"The Wave" shall be attended every day when the piece is turned on, to check for tear and wear from last night's performance and eventually change failed LED bars.

Space

The Wave needs the following type of space:

A field measured; 100m length x 10m width x 3.8m height.

The field must be flat with no obstacles, a uniform plan with a tolerance no more than $\pm 5\text{cm}$ throughout the whole length of The Wave.

During installation the space must be shutoff to the audience, and the container placed at the site during buildup.

Control center

In a radius of 40m from the center of The Wave we need to establish a control center.

The control center must be an indoor space and at least 6m^2 , with the possibility of cabling directly to the piece.

The travel-container for The Wave, can be used for the control center.

Power

One separate line of 230V with a standard red CEE 63 Ampere, 3 Phase (minimum CEE 32A, 3 Phase). The powerline must be at the entrance to the control center.

- We will provide an PDU, that splits the line from the control center to the piece. The entire piece uses max 2000W.



Other needs

Cable protection

Between the control-center and the center of The Wave, we need cable trays for 9 cables.

Shelter

Due to weather conditions we must have a sheltered area of at least 20m².

In addition, a portable tent must be available if it is raining during setup. This tent must be at least 4m high, 6m wide and 8m long.

Securing

Two moveable marquee blocks of at least 500kg in order of securing The Wave if necessary. And a pallet jack.

Personnel and setup-time estimate

4 persons from Vertigo are needed during a 5-day setup-time.

All the assistants just need to be in good shape, so they can lift the triangles. experience with cabling and stage rigging is a plus.

1. day - Foundation and general preparation

2-3 assistants needed in the afternoon.

- Unpacking the container, power setup, setup tent for assembling the triangles (2 from Vertigo)
- Outlaying foundation, placement of threaded steel rods, bolts and nuts (2-3 assistants)
- Foundation survey, point marking, leveling, laser measuring (2 from vertigo)

2. day - Triangle setup

6 assistants needed

- Preparation for cables and test setup, start of control center setup (2 from Vertigo)
- Triangle setup (2 from Vertigo and 6 assistants)
4 people for holding, setup and securing a triangle, meanwhile the other 4 is carrying out and preparing the next triangle. We should be able to put up $\frac{2}{3}$ of the gates.

3. day - Triangle setup and start cabling

6 assistants needed

Morning:

- Preparation for cables and test setup, setup control center (2 from Vertigo)
- Triangle setup, same procedure as the day before (2 from Vertigo and 6 assistants)

Afternoon:

- Cabling: power, ethernet and sound (1 from Vertigo and 4 assistants)
- Adjusting and leveling the triangles (1 from Vertigo and 2 assistants)

4. day - Cabling

3 assistants needed

- Control center setup, testing of the gates and fixing minor errors (2 from Vertigo)
- Cabling and securing cables with strips: power, ethernet and sound (1 from Vertigo and 3 assistants)



- Fixing hardware and light fixtures in the gates (1 from Vertigo)

5. day - Adjustments

2 assistants needed for the workshop

- Testing gates and contents (1 from Vertigo)
- Fixing hardware and light fixtures in the gates (2 from Vertigo)
- Workshop (1 from Vertigo and one or two assistants, being responsible for maintaining The Wave during exhibition period.)

Dismantling / taking down

1. day - Triangles and cables

5 assistants needed

- Taking down control center (1 from vertigo)
- Taking down triangles and cables (3 from Vertigo and 5 assistants)
4 people to take down, disassembling and caring away each triangle.

2. day - Foundation and control center

2 assistants needed

- Taking down control center and wrapping things up (2 from vertigo)
- Taking down foundation and tidying up the space (2 from Vertigo and 2 assistants)

Workshop

We need to train one or two assistants, for technical support and maintenance of the piece.

Training 12 hours or more depending on amount technical skills. Preferably after the installation has been setup and also that the trained has been a part of the crew during setup.

Woman or man, +21 years old, preferably a variety in ages between 21 – 60 and in physically good shape, so they are able to lift pieces of the gates and maintain electronics at ground-level.

The assistants need to be technically skilled in low current, assembly work, computer and network technology (Apple computer).

Transport

Standard 20' container measuring 6m long x 2.5m wide x 2.5m high.

Pickup at:

Det Kongelige Teater

Refshalevej 173A

1432 Copenhagen, Denmark

Contact:

Wolfgang +45 25 51 77 95

Insurance value

A total value of 1,100,000 DKK (Danish Kroner)

Contact

Vertigo ApS



Dannebrogsgade 53, krypt
1660 Copenhagen V

Frederik Svanholm +45 27 28 88 08

Mikkel Meyer +45 61 67 65 64

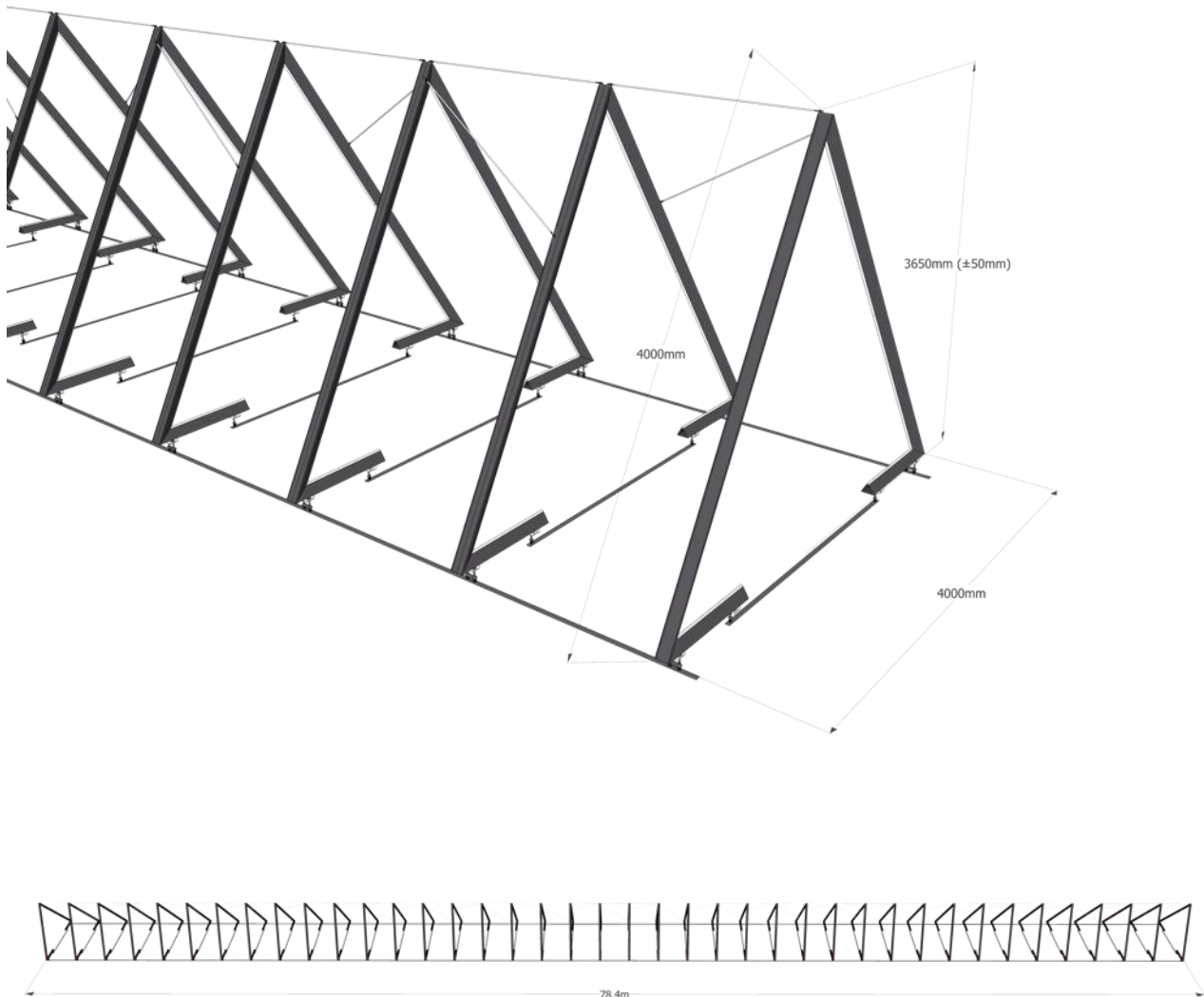
info@vertigo.dk www.vertigo.dk

Technical specs and info

Size

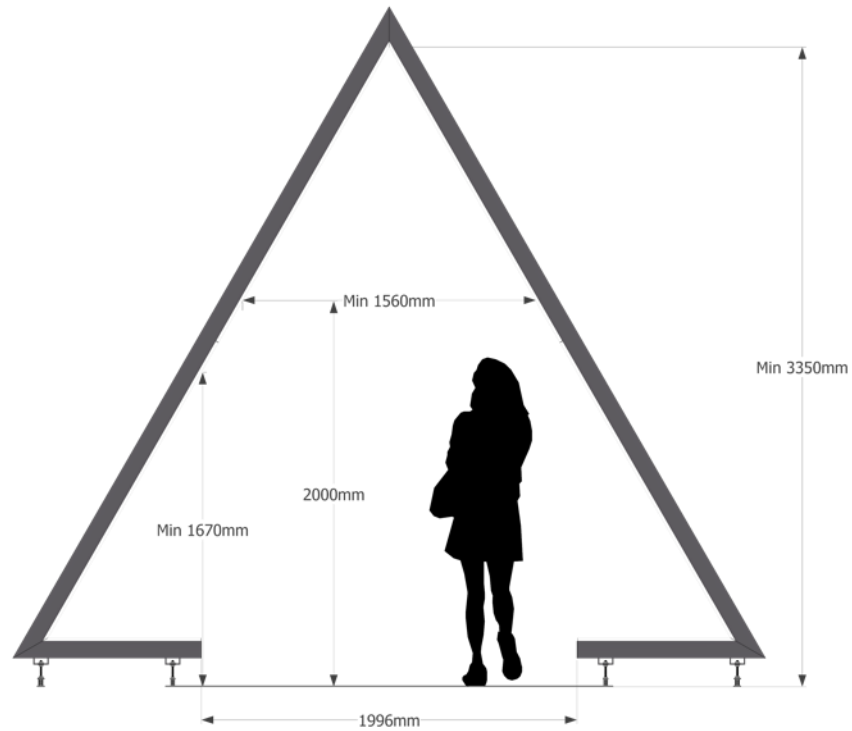
The Wave consists of 40 identical triangles.

The entire piece is 78.5m long, 4m wide and 3.7m high, and each gate weighs 74.35kg. The foundation weighs 276kg and the total weight of the piece is 3250kg.

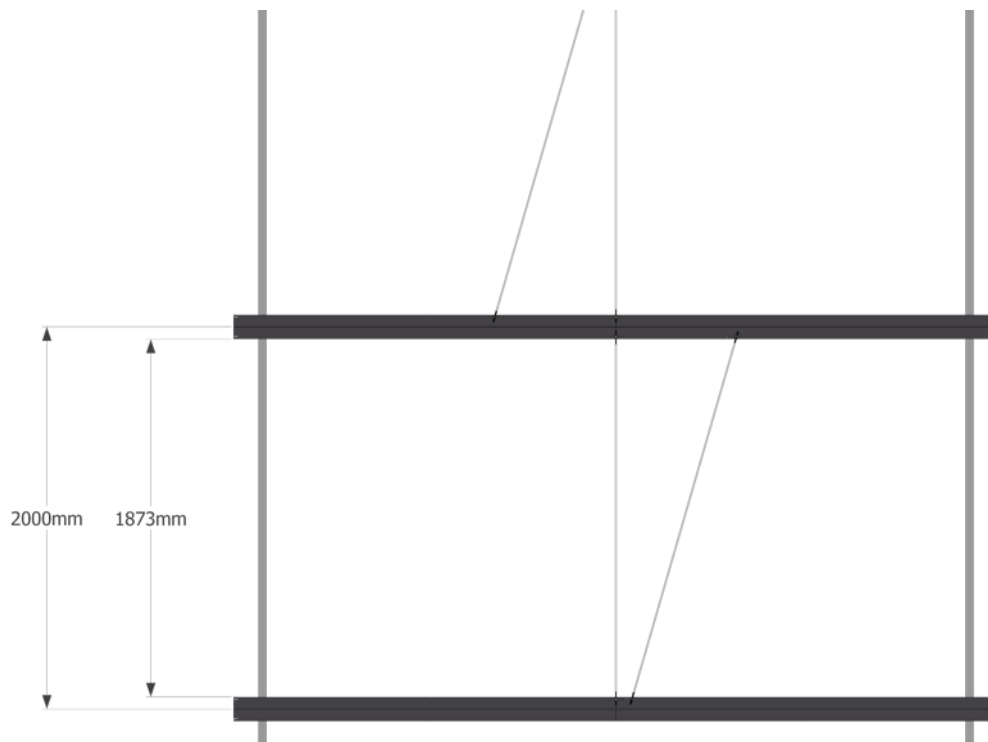




The space people can walk through in each triangle, is approximately 2m wide and 3m high.



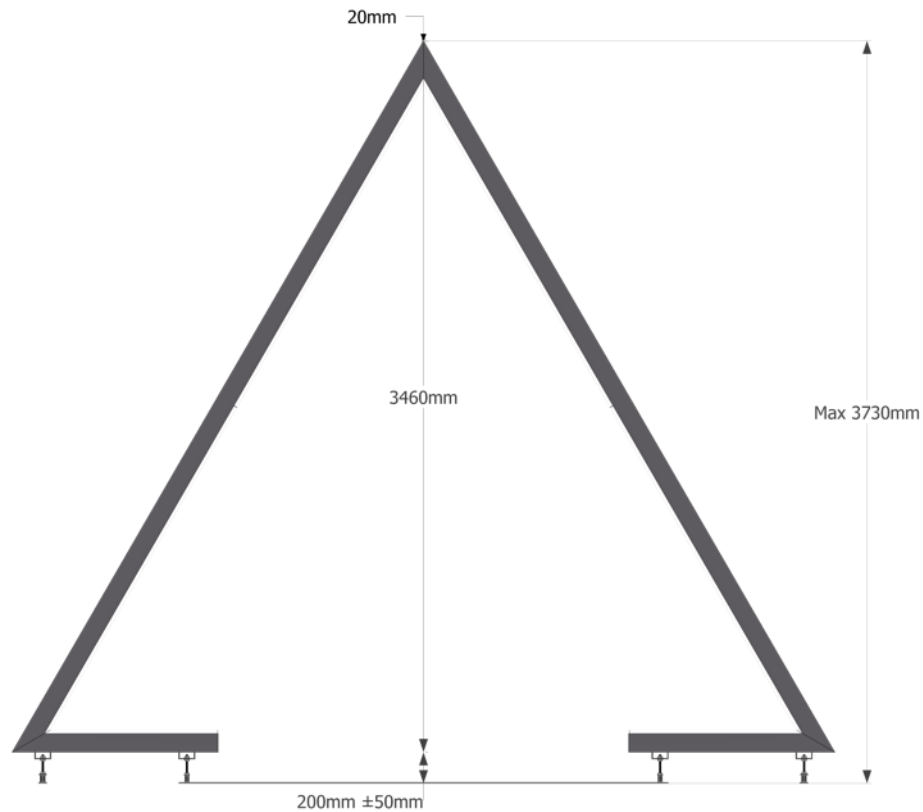
The distance between each triangle is 2m, where you can also walk through.



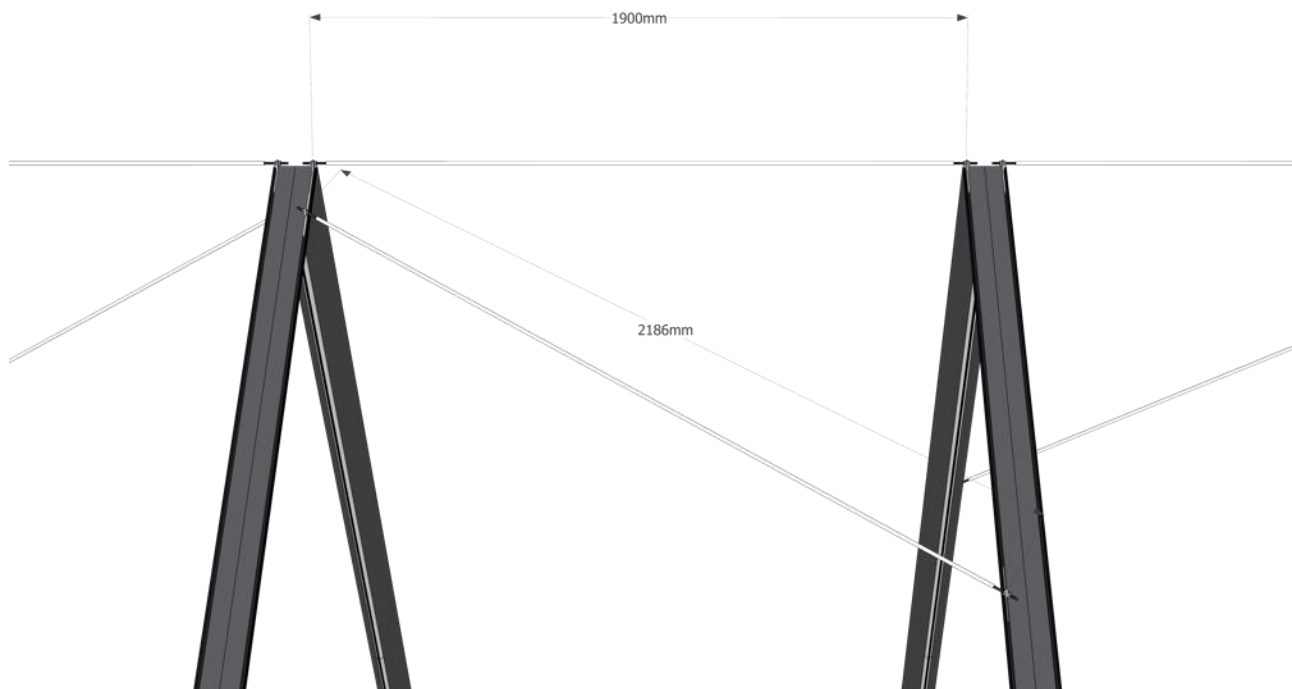


Structure

The triangle itself is exactly 3.46m high and 4m wide. It is a rectangular triangle, so each side is 4m. The triangles are lifted 200mm above the ground, with the ability to adjust it 50mm up and down.

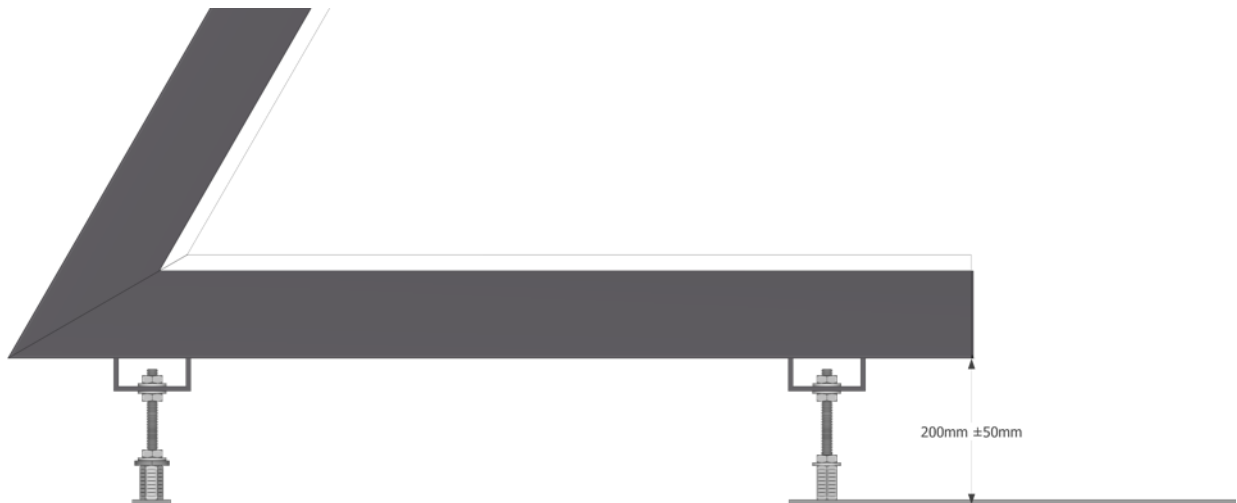


In the top between the triangles there are structural steel rods to ensure stability. One at the top and a diagonal at the side. The diagonal steel rods changes side between each triangle.

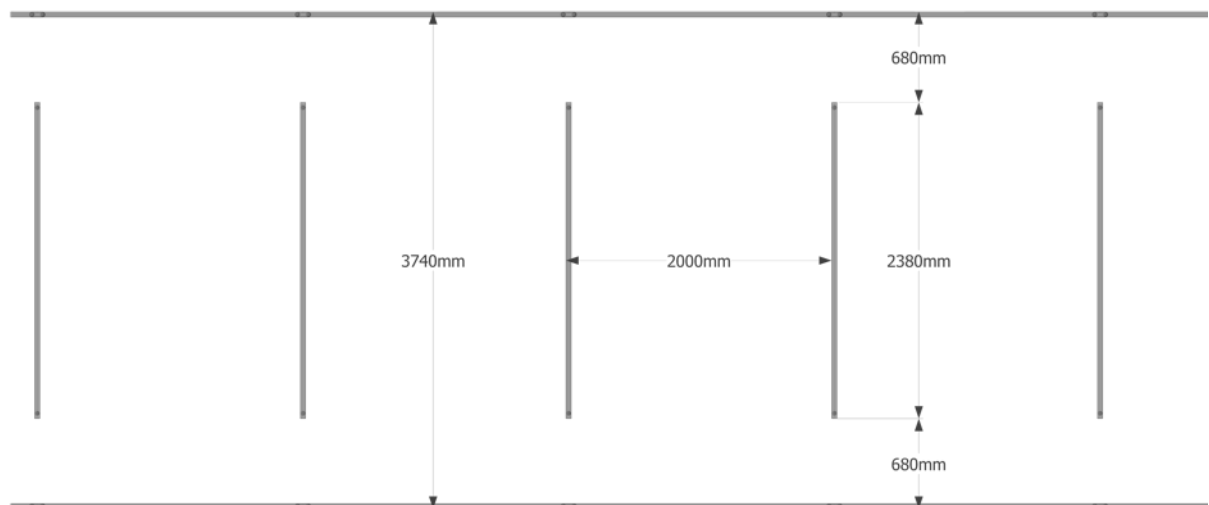




The triangle is attached to the foundation with steel threaded rods. The rods come in a variety of different lengths, so it is possible to level the entire piece with in a 100mm.



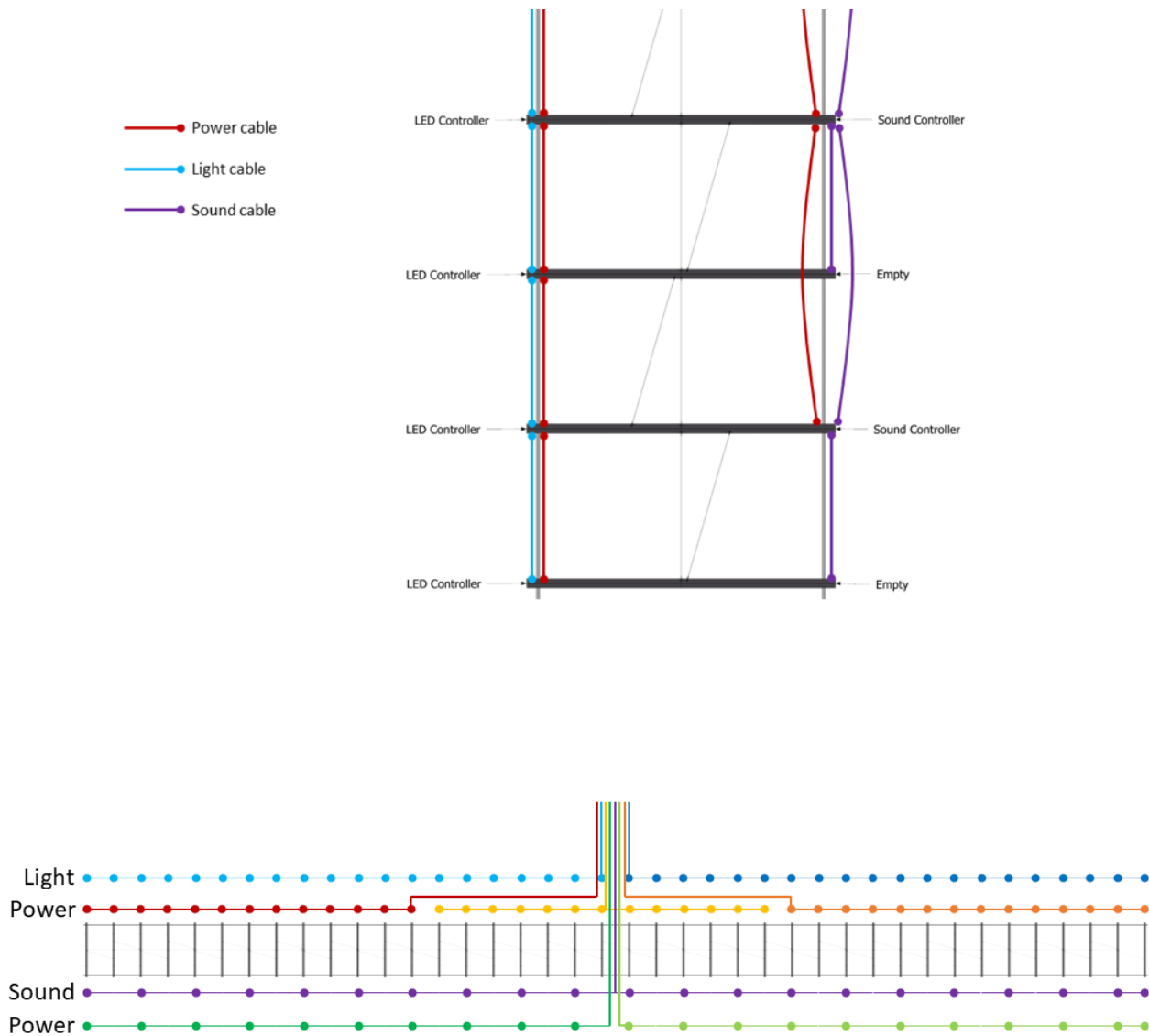
The foundation is made of steel bars. One bar between the two feet's and two bars in each side that goes between the gates.





Cabling

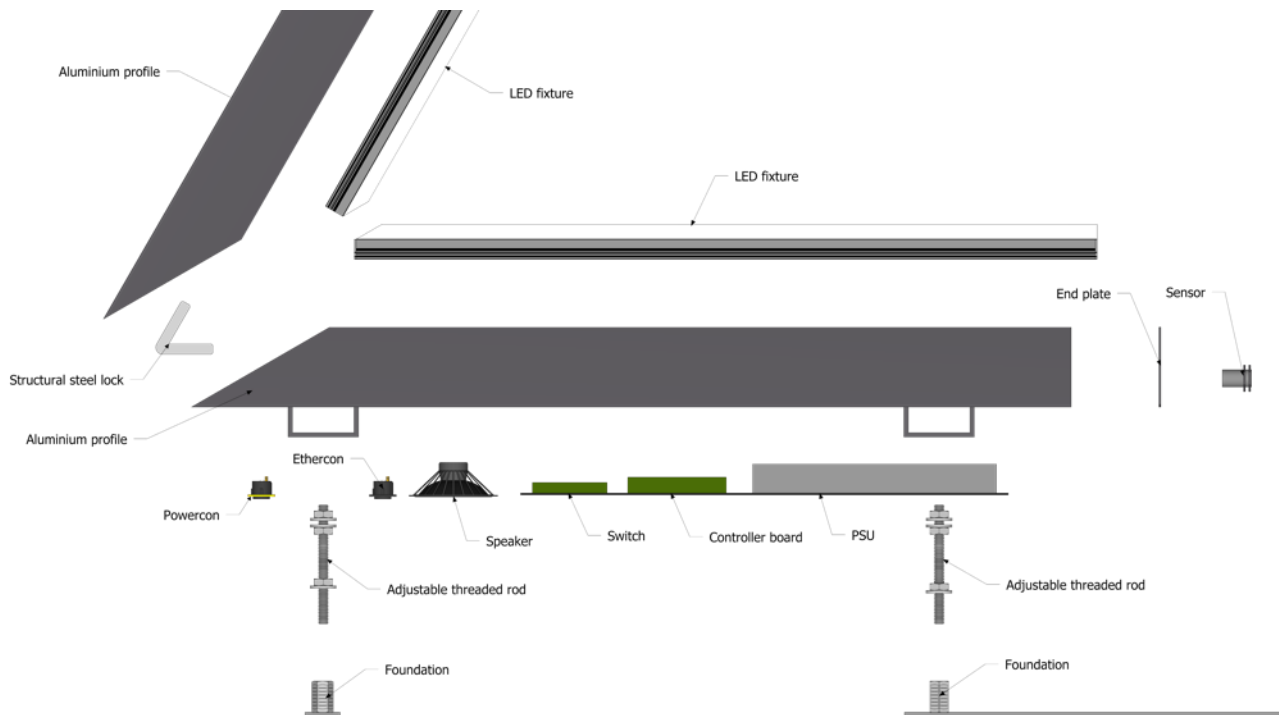
There are two sets of cables in each side of the piece. In the first side the LED controller is, here are both power and data cables running between each gate. In the second side the sound controller is, there are only one controller for every second gate, so the power and data cables are running between those, and there is a third cable running to the gates without controllers.



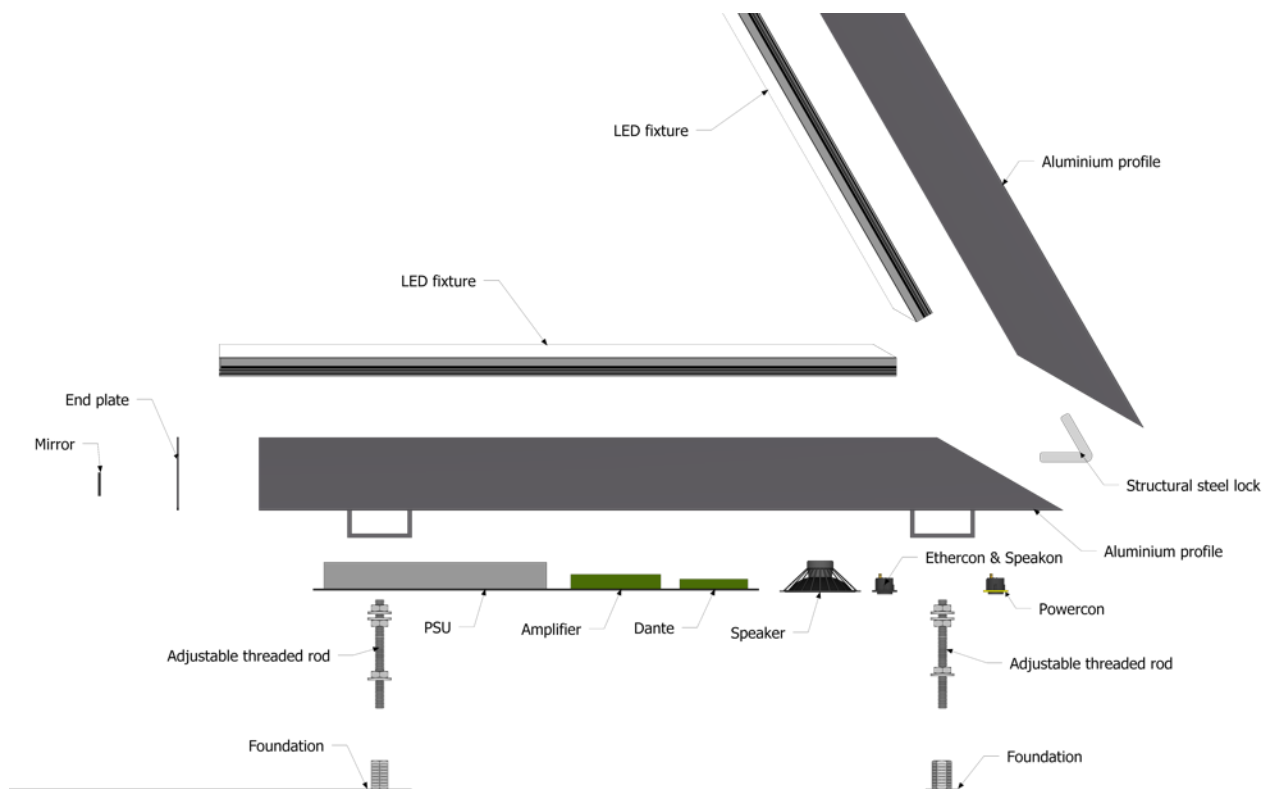


Design

1. foot (left side) is the controller for the LED fixtures and the sensor. Cables and bolts do not appear in the figure.

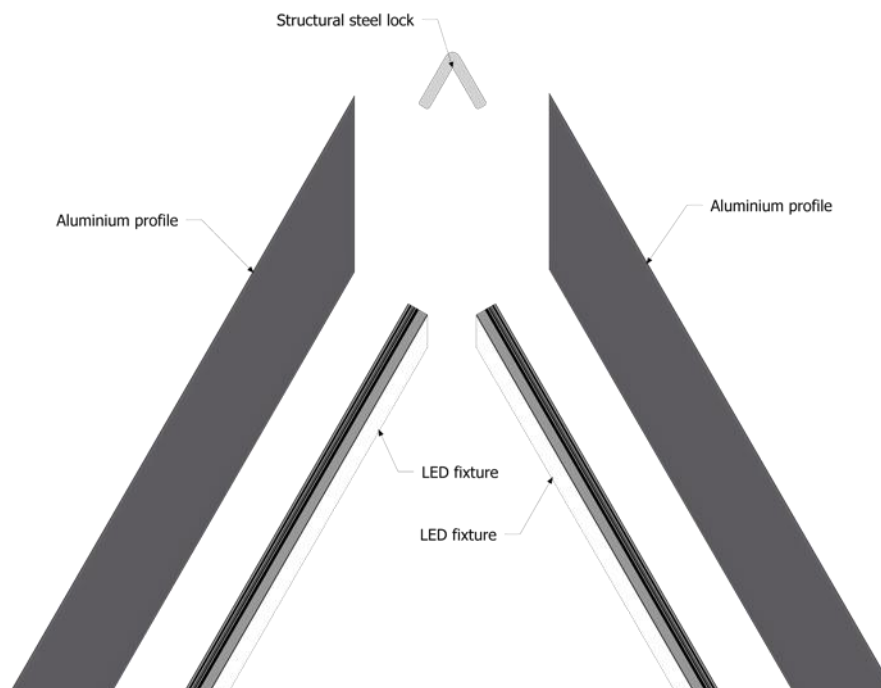


2. foot (right side) is the controller of the speakers. The right-side foot only has a controller for every second gate. Cables and bolts do not appear in the figure.





Top of the triangle. Cables and bolts do not appear in the figure.



Container

The entire piece comes in a standard 20 feet container, measuring: 6m long x 2.5m wide x 2.5m high. In the container is also the equipment needed for assembling, the main computer, the PDU and spare parts.

The total weight is approximately 5450-5950 kg.





Securing

The construction itself is structural secured in the foundation and the top steel rods. If the wind reaches above 13.8 m/s we recommend the structure to be rigged to the marquee blocks, placed 4m from each end of The Wave, with a 30kN spanset.